

The Ephemeris

March 2020

Volume 31 Number 01- The Official Publication of the San José Astronomical Association

March 2020 - June 2020 Events



Board & General Meetings

Saturday 3/7, 4/11, 5/9, 6/6

Board Meetings: 6 - 7:30pm

General Meetings: 7:30-9:30pm

Fix-It Day (2-4pm)

Sunday 3/15, 4/5, 5/3, 6/7

Solar Sunday (2-4pm)

Sunday 3/15, 4/5, 5/3, 6/7

Intro to the Night Sky Class & Houge Park 1Q In-Town Star Party

Friday 4/3, 5/1, 5/29, 6/26

Houge Park 3Q In-Town Star Party

Friday 3/13, 4/17, 5/15, 6/12

RCDO Starry Nights Star Party

Saturday 3/14, 4/18, 5/16, 6/13

Imaging SIG Mtg

Tuesday 3/17, 4/21, 5/19, 6/16

Astro Imaging Clinics

(at Little Uvas OSP)

Saturday 3/24, 6/20

Astro Imaging Workshops

(at Little Uvas OSP)

Saturday 4/22, 5/22

Binocular Star Gazing

Saturday (will resume in Spring 2020)

Please refer to the SJAA Website home page (www.sjaa.net) and click on any "Upcoming Event" highlighted in Blue. It will take you to the SJ-Astronomy Meetup page (www.meetup.com/SJ-Astronomy/events/) where you can find specific event times, locations, maps and possible cancellation due to weather.

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STARVIND
ASTRO PHOTOGRAPHY

Photo by SJAA Member: Arvind Hariharan
June 29, 2019
Location: Hollister, CA

SJAA Annual Awards and Board Elections

Credit: Abishek

The San Jose Astronomy Association held its annual awards and potluck dinner on February 8, 2020.

This time of the year we take a moment to appreciate the members who have shown significant contribution and support to the works and mission of SJAA. The board and all members value the fact that the recipients of these awards have put their invaluable time and energy into SJAA. In return they were honored with a certificate of appreciation and also a little astronomy related gift!

Kaushal Gangakhedkar:

About a year back, Kaushal took over the role of SJAA Membership Chair from Dave Ittner. He has made significant contributions towards maintaining the membership database. However, over the last few months, he went above and beyond the normal ask of the Membership Chair, to address the migration of our Yahoo observers group to the Google observers group, as well as creation of a new SJAA Members group. Kaushal manually downloaded nearly a thousand media files from the Yahoo group, to preserve a storehouse of multimedia information that had been shared in the Yahoo group over many years. He also reconciled the Yahoo group members against the membership DB and manually added members to both Google groups, as needed. Kaushal accomplished this Herculean task with a smile.



Left to Right: Treasurer Rob Jaworski, Vice-President Gerry Joyce and Kaushal G

Cat Cvengros: Cat stepped up to take over the formidable task of running the School Star Party (SSP) program that Jim Van Nuland ran for many decades. With able initial help from another member, Peter Mu, Cat put together a new mechanism to run the SSP, and implemented it. However, for over a year now, Cat has been single handedly running the program, which includes identifying available evenings for SSP, scheduling schools, working out the logistics with each school, securing volunteer astronomers for each event, monitoring the weather to determine if the party is a go or no-go, and frequently attending the parties herself with the SSP telescope. Unlike most other programs, the logistics for SSP need to be worked out multiple times a month during the season, frequently twice or three times a week. That involves devotion and significant time commitment.

Rob Jaworski: Rob Jaworski, our current Treasurer, has been serving SJAA in various capacities over the years. He was the President of SJAA for two terms a few years back. Once his term was over, he took over the role of the SJAA treasurer, and has been fulfilling this role ever since. He has also served as Secretary. But Rob's contribution is not just limited to being an officer of the club. Rob also volunteers to run the SJAA booth at the annual Cambrian festival, and is always willing to take on any other outreach events where SJAA has an opportunity to engage with the public. Rob also maintains SJAA's blog page as well as the Twitter feed. Rob's knowledge about SJAA history and dedication to the club is an inspiration to many of us.



Left to Right: Marianne Damon, Vice-President Gerry Joyce, and Treasure Rob Jaworski



Left to Right: Vice-President Gerry Joyce and Secretary Sandy Mohan

For the elections, Rob Jaworski, Sandy Mohan, Wolf Witt, and Glenn Newell were re-elected into their positions. These elections were for the Director position on the Board of Directors. Elections were followed by a delicious potluck with food and treats brought by SJAA members.

The San Jose Astronomical Association thanks **Cat Cvengros, Kaushal Gangakhedkar, and Rob Jaworski** for all the time, efforts and contributions to SJAA!! Congratulations!!

Strategic Meeting Summary

Credit: Swami Nigam



Standing Left to Right: Swami , Marianne, Gerry, Sandy, Vini, Bob T, Kanch, Peter, Rob J

Sitting Left to Right: Ken, Sukhada, Wolf and Natti

Not Pictured: Kaushal

SJAA leaders have started an exercise to evaluate and determine the strategic direction of the organization for the next few years. Then president, Swami Nigam, called together a meeting of SJAA directors, officers, and leaders for a brainstorming session on March 1 to kick off this effort.

As an initial step, the participants reaffirmed their support for the existing objective and purpose of the SJAA as described in Article 2 of the bylaws, "The primary objective and purpose of this corporation shall be to provide education about astronomy and related sciences to members and the public."

Additionally, the participants agreed that, as an organization we are doing well with our external constituency - the public - but that we could improve our performance with our internal constituency - our members. Towards that end we identified the following high level areas that we would like to focus resources on:

- Member education and advancement
- Engage proactively with new members
- Increase member participation in club operations and organization. Seek additional volunteers
- Social media and public relations strategy
- Online repository of Astronomy related materials
- Internship program for interested Bay Area students

In the coming weeks, we will be diving into details of each of these areas and will conduct more focused meetings. Small working groups will brainstorm each area and make recommendations of specific action plans to address each area. If you have ideas to contribute to this initiative and would like to join in the discussions, please email our president, Gerry Joyce (president@sjaa.net) or Swami Nigam (director5@sjaa.net)

Clear Skies Bring Out the Amateur Astronomers in the Bay Area - SJAA coverage on KCBS Radio Show by Keith Menconi

Credits: Keith Manconi and KCBS



SJAA tweets about the KCBS interview of Swami Nigam by Keith Marconi

The San Jose Astronomy Association was covered as part of the KCBS Radio Show by Keith Menconi and the following is an edited excerpt of the transcript from the broadcast.

As the winter storms clear up, bay area amateur astronomers are cracking out the telescopes once again and hitting the streets for the first urban star parties of the new year. KCBS reporter Keith Menconi has more from San Jose.

Keith Menconi - It turns out, star gazing is pretty hard to do when there are clouds in the way.

Swami Nigam - Yes, visual astronomy, but not during cloudy nights.

Keith Menconi - So San Jose Astronomical Association President Swami Nigam says, for the past few winter months, his group has been unable to gather for the regularly scheduled stargazing meetups.

Swami Nigam - Most of our events had to be cancelled just because of the weather.

Keith Menconi - Now that they can get together, the relief is palpable.

Guest observer's remark during an SJAA star party - "That is amazing!"

Keith Menconi - This star party is in a San Jose parking lot. The light pollution may not be ideal, but the central location gives an opportunity for more newcomers to join in. The goal - spreading the stargazing joy, far and wide!

Swami Nigam - Yes, exactly! It is a lot about education, we have a wide variety of experience among the astronomers who come here.

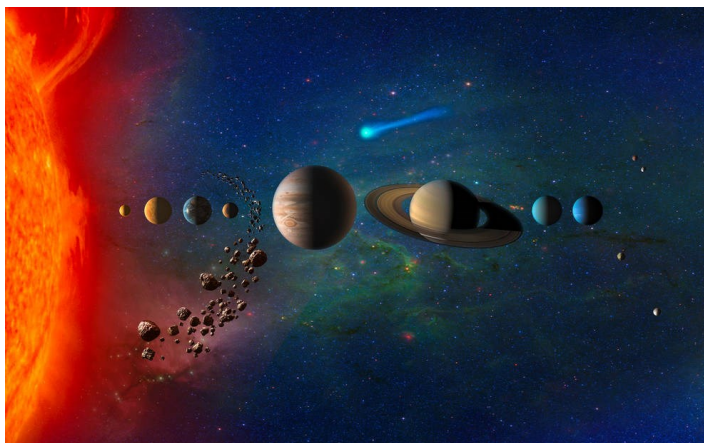
SJAA member showing a guest observer - "The brightest one in the middle, pick that one out and ..."

Keith Menconi - This gathering of a few dozen is just the beginning. Nigam says the numbers really pick up when it starts to get warm. In San Jose, Keith Menconi, KCBS.

To listen to the audio version of the interview please go to: <https://omny.fm/shows/kcbsam-on-demand/clear-skies-bring-out-the-amateur-astronomers-in-t>

NASA Selects Four Possible Missions to Study the Secrets of the Solar System

Credits: NASA



Artist concept of the solar system.
Credits: NASA

NASA has selected four Discovery Program investigations to develop concept studies for new missions. Although they're not official missions yet and some ultimately may not be chosen to move forward, the selections focus on compelling targets and science that are not covered by NASA's active missions or recent selections. Final selections will be made next year.

NASA's Discovery Program invites scientists and engineers to assemble a team to design exciting planetary science missions that deepen what we know about the solar system and our place in it. These missions will provide frequent flight opportunities for focused planetary science investigations. The goal of the program is to address pressing questions in planetary science and increase our understanding of our solar system.

"These selected missions have the potential to transform our understanding of some of the solar system's most active and complex worlds," said Thomas Zurbuchen, associate administrator of NASA's Science Mission Directorate. "Exploring any one of these celestial bodies will help unlock the secrets of how it, and others like it, came to be in the cosmos."

Each of the four nine-month studies will receive \$3 million to develop and mature concepts and will conclude with a Concept Study Report. After evaluating the concept studies, NASA will continue development of up to two missions towards flight.

The proposals were chosen based on their potential science value and feasibility of development plans following a competitive peer-review process.

The selected proposals are:

DAVINCI+ (Deep Atmosphere Venus Investigation of Noble gases, Chemistry, and Imaging Plus)

DAVINCI+ will analyze Venus' atmosphere to understand how it formed, evolved and determine whether Venus ever had an ocean. DAVINCI+ plunges through Venus' inhospitable atmosphere to precisely measure its composition down to the surface. The instruments are encapsulated within a purpose-built descent sphere to protect them from the intense environment of Venus. The "+" in DAVINCI+ refers to the imaging component of the mission, which includes cameras on the descent sphere and orbiter designed to map surface rock-type. The last U.S.-led, in-situ mission to Venus was in 1978. The results from DAVINCI+ have the potential to reshape our understanding of terrestrial planet formation in our solar system and beyond. James Garvin of NASA's Goddard Space Flight Center in Greenbelt, Maryland, is the principal investigator. Goddard would provide project management.

Io Volcano Observer (IVO)

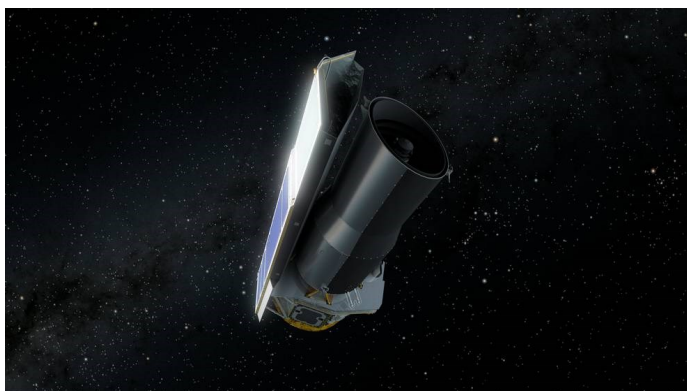
IVO would explore Jupiter's moon, Io, to learn how tidal forces shape planetary bodies. Io is heated by the constant crush of Jupiter's gravity and is the most volcanically active body in the solar system. Little is known about Io's specific characteristics, such as whether a magma ocean exists in its interior. Using close-in flybys, IVO would assess how magma is generated and erupted on Io. The mission's results could revolutionize our understanding of the formation and evolution of rocky, terrestrial bodies, as well as icy ocean worlds in our solar system, and extrasolar planets across the universe. Alfred McEwen of the University of Arizona in Tucson is the principal investigator. The Johns Hopkins University Applied Physics Laboratory in Laurel, Maryland would provide project management.

TRIDENT

Trident would explore Triton, a unique and highly active icy moon of Neptune, to understand pathways to habitable worlds at tremendous distances from the Sun. NASA's Voyager 2 mission showed that Triton has active resurfacing—generating the second youngest surface in the solar system—with the potential for erupting plumes and an atmosphere. Coupled with an ionosphere that can create organic snow and the potential for an interior ocean, Triton is an exciting exploration target to understand how habitable worlds may develop in our solar system and others. Using a single fly-by, Trident would map Triton, characterize active processes, and determine whether the predicted subsurface ocean exists. Louise Prockter of the Lunar and Planetary Institute/Universities Space Research Association in Houston is the principal investigator. NASA's Jet Propulsion Laboratory (JPL) in Pasadena, California, would provide project management.

NASA's Spitzer Space Telescope Ends Mission of Astronomical Discovery

Credit: Nasa and JPL



NASA's Spitzer Space Telescope has concluded after more than 16 years of exploring the universe in infrared light.

Credits: NASA/JPL-Caltech

After more than 16 years studying the universe in infrared light, revealing new wonders in our solar system, our galaxy, and beyond, NASA's Spitzer Space Telescope's mission has come to an end.

Mission engineers confirmed at 2:30 p.m. PST (5:30 p.m. EST) Thursday the spacecraft was placed in safe mode, ceasing all science operations. After the decommissioning was confirmed, Spitzer Project Manager Joseph Hunt declared the mission had officially ended.

Launched in 2003, Spitzer was one of NASA's four Great Observatories, along with the Hubble Space Telescope, the Chandra X-ray Observatory and the Compton Gamma Ray Observatory. The Great Observatories program demonstrated the power of using different wavelengths of light to create a fuller picture of the universe.

"Spitzer has taught us about entirely new aspects of the cosmos and taken us many steps further in understanding how the universe works, addressing questions about our origins, and whether or not we are alone," said Thomas Zurbuchen, associate administrator of NASA's Science Mission Directorate in Washington. "This Great Observatory has also identified some important and new questions and tantalizing objects for further study, mapping a path for future investigations to follow. Its immense impact on science certainly will last well beyond the end of its mission."

Among its many scientific contributions, Spitzer studied comets and asteroids in our own solar system and found a previously unidentified ring around Saturn. It studied star and planet formation, the evolution of galaxies from the ancient universe to today, and the composition of interstellar dust. It also proved to be a powerful tool for

detecting exoplanets and characterizing their atmospheres. Spitzer's best-known work may be detecting the seven Earth-size planets in the TRAPPIST-1 system – the largest number of terrestrial planets ever found orbiting a single star – and determining their masses and densities.

In 2016, following a review of operating astrophysics missions, NASA made a decision to close out the Spitzer mission in 2018 in anticipation of the launch of the James Webb Space Telescope, which also will observe the universe in infrared light. When Webb's launch was postponed, Spitzer was granted an extension to continue operations until this year. This gave Spitzer additional time to continue producing transformative science, including insights that will pave the way for Webb, which is scheduled to launch in 2021.

"Everyone who has worked on this mission should be extremely proud today," Hunt said. "There are literally hundreds of people who contributed directly to Spitzer's success, and thousands who used its scientific capabilities to explore the universe. We leave behind a powerful scientific and technological legacy."



Spitzer Project Manager Joseph Hunt stands in Mission Control at NASA's Jet Propulsion Laboratory in Pasadena, California, on Jan. 30, 2020, declaring the spacecraft decommissioned and the Spitzer mission concluded.

Credits: NASA/JPL-Caltech



Date: December 30, 2019

Location: Sierra Nevada, Yosemite

Imaging Telescope: Celestron RASA

Imaging Camera: QHY168C

Mount: Software Bisque Paramount MX+

Software: StarTools, TheSkyX Pro, SkyTools, SharpCap 3.1 Sharp Cap, PixInsight1.8, PHD Guiding 2 PHD 2 Guiding



Dates: Feb 10, 11 2020

Location: Mountain View, Ca

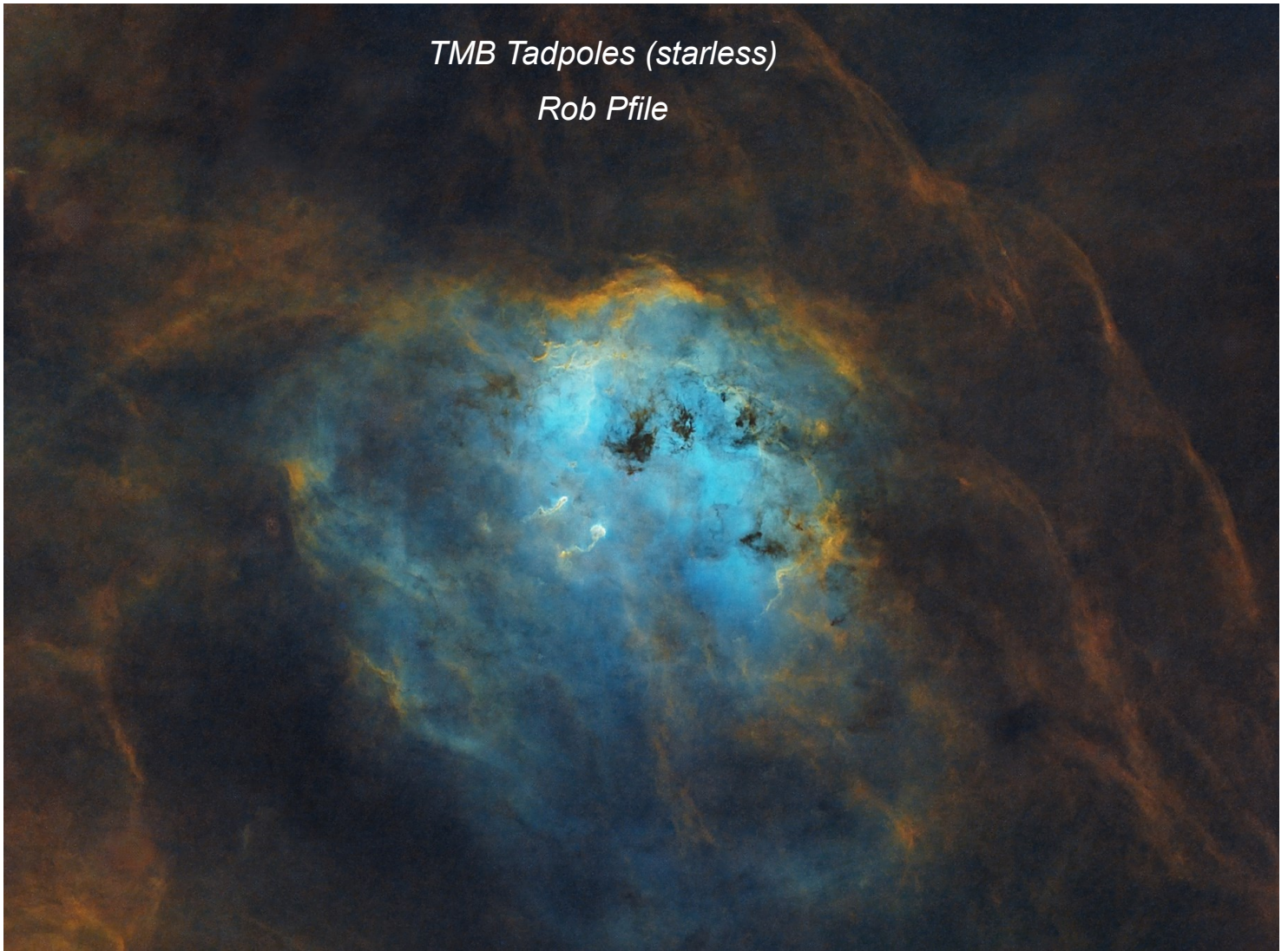
Telescope: Astro-Tech AT130EDT

Camera: Nikon D5500 (Ha mod)

Mount: iOptron iEQ45 Pro

Software: Pleiades Astrophoto PixInsight 1.8 Ripely

SJAA Member Astrophoto Gallery



Date: Nov 1, 2018

Location: Oakland, CA

Telescope: TMB-92SS

Camera: SBIG STT 8300M

Mount: Astro-Physics Mach1 GTO

Software: PHD guiding, PixInsight, Main Sequence Software Sequence Generator Pro, Nikita Misiura Starnet ++

SJAA Member Astrophoto Gallery

Helix Nebula
Swaroop Shere



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Date: August 29, 2019

Location: Stellar Skies Observatory, Pontotoc, TX

Telescope: Orion 10" f/3.9 Astrograph

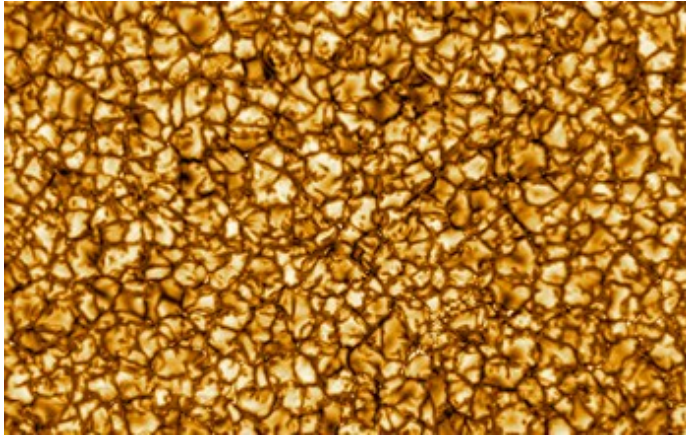
Camera: QSI683 wsg-8

Mount: Starlight Express Lodestar X2

Software: Software Bisque Sky X Pro, PixInsight 1.8 PixInsight , Sequence Generator Pro, PHD2 Guiding, Diffraction Limited MaxIm DL Pro 6

NSF's newest solar telescope produces first images

Credit NSF



The Daniel K. Inouye Solar Telescope has produced the highest resolution image of the sun's surface ever taken. In this picture, taken at 789 nanometers (nm), we can see features as small as 30km (18 miles) in size for the first time ever. The image shows a pattern of turbulent, "boiling" gas that covers the entire sun. The cell-like structures -- each about the size of Texas -- are the signature of violent motions that transport heat from the inside of the sun to its surface. Hot solar material (plasma) rises in the bright centers of "cells," cools off and then sinks below the surface in dark lanes in a process known as convection. In these dark lanes we can also see the tiny, bright markers of magnetic fields. Never before seen to this clarity, these bright specks are thought to channel energy up into the outer layers of the solar atmosphere called the corona. These bright spots may be at the core of why the solar corona is more than a million degrees.

Credit: NSO/AURA/NSF

The first images from NSF's Inouye Solar Telescope show a close-up view of the sun's surface, which can provide important detail for scientists. The images show a pattern of turbulent "boiling" plasma that covers the entire sun. The cell-like structures -- each about the size of Texas -- are the signature of violent motions that transport heat from the inside of the sun to its surface. That hot solar plasma rises in the bright centers of "cells," cools, then sinks below the surface in dark lanes in a process known as convection.

"Since NSF began work on this ground-based telescope, we have eagerly awaited the first images," said France Córdova, NSF director. "We can now share these images and videos, which are the most detailed of our sun to date. NSF's Inouye Solar Telescope will be able to map the magnetic fields within the sun's corona, where solar eruptions occur that can impact life on Earth. This telescope will improve our understanding of what drives space weather and ultimately help forecasters better predict solar storms."

Expanding knowledge

The sun is our nearest star -- a gigantic nuclear reactor that burns about 5 million tons of hydrogen fuel every second. It has been doing so for about 5 billion years and will continue for the other 4.5 billion years of its lifetime. All that energy radiates into space in every direction, and the tiny fraction that hits Earth makes life possible. In the 1950s, scientists figured out that a solar wind blows from the sun to the edges of the solar system. They also concluded for the first time that we live inside the atmosphere of this star. But many of the sun's most vital processes continue to confound scientists.

"On Earth, we can predict if it is going to rain pretty much anywhere in the world very accurately, and space weather just isn't there yet," said Matt Mountain, president of the Association of Universities for Research in Astronomy, which manages the Inouye Solar Telescope. "Our predictions lag behind terrestrial weather by 50 years, if not more. What we need is to grasp the underlying physics behind space weather, and this starts at the sun, which is what the Inouye Solar Telescope will study over the next decades."

January 29, 2020

Just released first images from the National Science Foundation's Daniel K. Inouye Solar Telescope reveal unprecedented detail of the sun's surface and preview the world-class products to come from this preeminent 4-meter solar telescope. NSF's Inouye Solar Telescope, on the summit of Haleakala, Maui, in Hawai'i, will enable a new era of solar science and a leap forward in understanding the sun and its impacts on our planet.

Activity on the sun, known as space weather, can affect systems on Earth. Magnetic eruptions on the sun can impact air travel, disrupt satellite communications and bring down power grids, causing long-lasting blackouts and disabling technologies such as GPS.

The motions of the sun's plasma constantly twist and tangle solar magnetic fields. Twisted magnetic fields can lead to solar storms that can negatively affect our technology-dependent modern lifestyles. During 2017's Hurricane Irma, the National Oceanic and Atmospheric Administration reported that a simultaneous space weather event brought down radio communications used by first responders, aviation and maritime channels for eight hours on the day the hurricane made landfall.

Finally resolving these tiny magnetic features is central to what makes the Inouye Solar Telescope unique. It can measure and characterize the sun's magnetic field in more detail than ever seen before and determine the causes of potentially harmful solar activity.

"It's all about the magnetic field," said Thomas Rimmele, director of the Inouye Solar Telescope. "To unravel the sun's biggest mysteries, we have to not only be able to clearly see these tiny structures from 93 million miles away but very precisely measure their magnetic field strength and direction near the surface and trace the field as it extends out into the million-degree corona, the outer atmosphere of the sun."

Better understanding the origins of potential disasters will enable governments and utilities to better prepare for inevitable future space weather events. It is expected that notification of potential impacts could occur earlier -- as much as 48 hours ahead of time instead of the current standard, which is about 48 minutes. This would allow more time to secure power grids and critical infrastructure and to put satellites into safe mode.

The engineering

To achieve the proposed science, this telescope required important new approaches to its construction and engineering. Built by NSF's National Solar Observatory and managed by AURA, the Inouye Solar Telescope combines a 13-foot (4-meter) mirror -- the world's largest for a solar telescope -- with unparalleled viewing conditions at the 10,000-foot Haleakala summit.

Focusing 13 kilowatts of solar power generates enormous amounts of heat -- heat that must be contained or removed. A specialized cooling system provides crucial heat protection for the telescope and its optics. More than seven miles of piping distribute coolant throughout the observatory, partially chilled by ice created on site during the night.

The dome enclosing the telescope is covered by thin cooling plates that stabilize the temperature around the telescope, helped by shutters within the dome that provide shade and air circulation. The "heat-stop" (a high-tech, liquid-cooled, doughnut-shaped metal) blocks most of the sunlight's energy from the main mirror, allowing scientists to study specific regions of the sun with unparalleled clarity.

The telescope also uses state-of-the-art adaptive optics to compensate for blurring created by Earth's atmosphere. The design of the optics ("off-axis" mirror placement) reduces bright, scattered light for better viewing and is complemented by a cutting-edge system to precisely focus the telescope and eliminate distortions created by the Earth's atmosphere. This system is the most advanced solar application to date.

"With the largest aperture of any solar telescope, its unique design, and state-of-the-art instrumentation, the Inouye Solar Telescope -- for the first time -- will be able to perform the most challenging measurements of the sun," Rimmele said. "After more than 20 years of work by a large team devoted to designing and building a premier solar research observatory, we are close to the finish line. I'm extremely excited to be positioned to observe the first sunspots of the new solar cycle just now ramping up with this incredible telescope."

New era of solar astronomy

NSF's new ground-based Inouye Solar Telescope will work with space-based solar observation tools such as NASA's Parker Solar Probe (currently in orbit around the sun) and the European Space Agency/NASA Solar Orbiter (soon to be launched). The three solar observation initiatives will expand the frontiers of solar research and improve scientists' ability to predict space weather.



Kid Spot Jokes:

- ♦ **Why couldn't the astronaut book a room on the moon?**
Because it was full
- ♦ **Why does NASA believe there might be life on Mars?**
The CD player was stolen from their Mars Rover.
- ♦ **How many astronomers does it take to change a light bulb ?**
None, astronomers aren't scared of the dark.

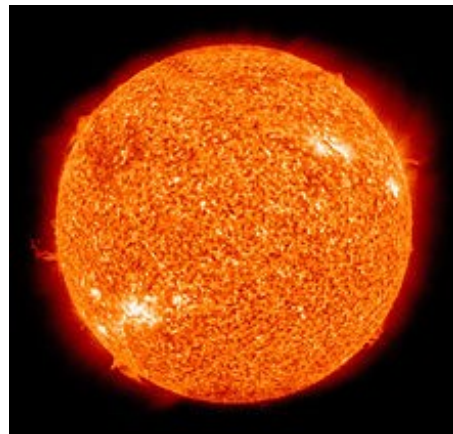
Fun Facts

The Mars Rover 'Opportunity'



Credits: NASA/JPL/Cornell University

Our Sun



Credits: NASA/SDO (AIA)

- * Opportunity explored the Red Planet for more than 14 years, which was 55 times its designed lifespan! The end of its mission was declared only in February 2019, exactly 1 year ago.
- * Opportunity is nicknamed 'Oppy!'
- * Sadly, 'Oppy' lost its twin sibling 'Spirit' long ago in 2010 when it got stuck in a sand trap on the other side of Mars!
- * Oppy also holds the record for longest distance traveled on the surface of another world—45.16 kilometers or 28.06 miles! (a little over 26,219 mile full marathon)

- * Only one two-billionth of the Sun's energy hits the Earth! Imagine if it all did!
- * In 2020, we have the most detailed image ever taken of the Sun by the Daniel K. Inouye Solar Telescope that shows plasma cells the size of Texas!
- * The mass of Mercury + Venus + Earth + Mars + Jupiter + Saturn + Uranus + Neptune + any other non-planetary objects including the millions of asteroids, meteors etc. only constitute 0.14% the mass of the solar system. Just our sun is the remaining 99.86%!
- * The interior of the sun is estimated to be 15,000,000 degrees Celsius!
- * It takes 5 hours or 18000 sec to get from San Jose to Los Angeles by car. Imagine if you could get to Los Angeles in 2 seconds! That's about how fast the sun travels in the milky way galaxy!

From the Board of Directors

Announcements

- ◇ San Jose Astronomical Association is monitoring closely the coronavirus, COVID-19, situation in Santa Clara county. In accordance with advisement from Santa Clara County Health Department we may be canceling some events in the next few weeks, as appropriate - especially those that involve close contact and shared surface on telescopes. Please check our Meetup to determine if an event has been canceled before heading out to the event. The board will periodically review the status of the outbreak and the guidance from the Santa Clara county to determine the impact upon upcoming events.
- ◇ Rob Jaworski, Sandy Mohan, Wolf Witt, and Glenn Newell were re-elected into their respective Board of Director positions
- ◇ Kanch to take over the In-Town Star Party coordinator
- ◇ Abhishek Ramasubramanian has been elected as the new Ephemeris Content Editor
- ◇ Officer elections will be held during the March board meeting scheduled for March 7, 2020.
- ◇ New Open Positions

Board Meeting Excerpts

December 14, 2019

In attendance

Swami, Sandy, Glenn, Ken, Gerry, Peter Wolf
Excused Absences: Wolf, Vini, Sukhada, Rob
Guests: Marianne

Migration of Google groups

All current members have been migrated from Yahoo to SJAA Observers group. All pictures posted on Yahoo observers group will be deleted as of

Dec 14, 2019. Photos have been saved and will be made available to SJAA members

Strategic Meeting

A special meeting to discuss some thoughts and ideas about the strategic direction of the club. Summary of the meeting held on March 1st is included on page 3 of this issue.

January 11, 2020

In attendance

Swami, Sandy, Vini, Glenn, Sukhada, Wolf, Vini, Rob J, Ken, Gerry, Peter
Excused Absences: none
Guests: Emi, Dave, Marianne

Upcoming Board Election

Board elections are coming up in February and 4 of the 9 board positions are up for elections.

Nominations for annual awards

Nominations are invited for the annual SJAA recognition awards.

Ephemeris Content Editor position

Abishek has expressed interest in taking the place. Vote on this next meeting.

February 8, 2020

In attendance

Swami, Wolf, Vini, Rob J, Glenn, Sandy, Ken
Excused Absences: Swami, Peter
Guests: Marianne, Emi, Nick, Paul, Kaushal, Abishek

KCBS Interview with Swami

Mary Hughes & Keith Menconi came to ITSP and also observed the Intro to Astro class. Swami & Wolf met with them and talked about SJAA. The interview aired recently. It was very good. For the interview please see page 4 of this issue.

Ephemeris Content Editor

Swami nominated Abishek Ramasubramanian for the position of Ephemeris Content Editor. It was approved and he is the new Content Editor.

New Open Positions

New positions to recruit for: SJAA Liaison, Volunteer Chair

Officer Elections

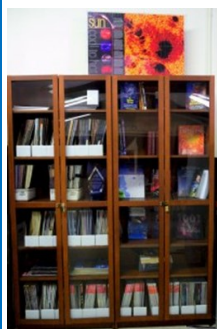
Officer elections for President, Vice-President, Treasurer, and Secretary positions will be held during the March board meeting. Any SJAA member in good standing that is interested in running for an officer seat is welcome to come to the board meeting on 7 March 2020. Please see the organization's by-laws for details.

<https://www.sjaa.net/about/club-by-laws/#A4>

Officer Elections

Officer elections for President, Vice-President, Treasurer, and Secretary were held during the March board meeting. Rob Jaworski was re-elected as Treasurer. Gerry Joyce was elected as President replacing Swami Nigam after his 2 terms passed. Ken Miura was elected as Vice-President replacing Gerry Joyce. Emi Nikolov was elected as Secretary replacing Sandy Mohan who will still serve as a board member.

SJAA Library



SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers! You may still send all your questions/comments

to librarian@sjaa.net.

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. Headed up by Vini Carter, the Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

<http://www.sjaa.net/programs/fixit/>

Solar Observing

Solar observing sessions, headed up by Wolf Witt, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change.

<http://www.meetup.com/SJ-Astronomy/>

Quick START Program

The Quick START Program, headed up by Dave Ittner, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick START program is undergoing a redesign to make it more accessible to SJAA members, so signups are currently on hold.

<http://www.sjaa.net/programs/quick-start/>

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly session, run by David Grover, each with a similar format, with only the content changing to reflect what's currently in the night

sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with views through SJAA Telescopes at the In-Town Star Party, to get a better look at the night's celestial objects.

<http://www.sjaa.net/programs/beginners-astronomy/>

Loaner Program

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met. If you are an SJAA member and an experienced observer or have been through the SJAA Quick START program please fill this form to request a particular item. Please also consider donating unused equipment.

<http://www.sjaa.net/programs/loaner-telescope-program/>

Astro Imaging Special Interest Group (SIG)

SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. Run by Bruce Braunstein, the Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

<http://www.sjaa.net/programs/imaging-sig/>

Astro Imaging Workshops and Field Clinics

Not to be confused with the SIG group this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astro-photography, to have a chance of seeing what it is all about.

Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact SJAA's (Program Coordinator) for

additional information.

<http://www.sjaa.net/programs/school-star-party/>

SJAA Contacts

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SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles, including Observation Reports for publication should be submitted by no later than the 20th of the month of February, May, August and November.

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Questions? Send e-mail to: memberships@sjaa.net

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at: <http://www.sjaa.net/join-the-sjaa/>

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